

# Smoothing Surgical Bed Census via Block Scheduling

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# 1 Introduction

Operation research in healthcare has been receiving more and more attention with the decreasing birth rate and the uprising of the global average longevity [1]. One major issue for hospitals is the inefficiency of bed allocation. A patient that requires a surgical operation will at least visit three place at the hospital: an operating room (OR), the post anesthesia care unit (PACU), and then a surgical inpatient bed. Different patients have different conditions and thus have different dates of surgery. This raises an issue because the number of surgeons and beds are limited, and the hospital may end up having 100 beds occupied on Wednesday but only 20 occupied beds on Friday. If we can change the days that some people have their surgical operation, then we can improve the flow by smoothing out the bed occupancy. In this study, we applied a mixed integer programming (MIP) model to determine a weekly operating room allocation schedule, to reduce the weekly peak in bed occupancy.

This study is inspired by the paper “*An Optimization Framework for Smoothing Surgical Bed Census via Strategic Block Scheduling*” by Carnes et al. (2011) [2]. The paper introduces a generalizable optimization framework to tackle the problem of peak bed utilization in hospitals and applied this approach to an academic hospital, Massachusetts General Hospital (MGH), which has 52 ORs each day and about 38,000 annual surgical operations. The framework employs integer programming with the objective of minimizing the maximum average bed census across all inpatient units. This framework is general enough to be applied to many other hospitals [2]. Our work in this paper replicates the methods outlined in the paper and extend them to similar bed-allocation problems using alternative datasets and modified constraints.

In the hospital, a surgeon can selectively schedule a patient based on the surgeon’s availability. Having ownership of a operation room (OR) allows the surgeon access to scheduling cases to be performed in that OR on that specified date exclusively. A monthly repeating block schedule is designed to show who has the ownership of each OR on each day. An occupied block is often released as the date draws nearer so that other surgeons may schedule cases in the unused time of the OR. By strategically rearranging the block schedule (what days of the week a particular surgeon has access to an OR), we can effectively smooth the peak of patient bed usage, and thus increase the bed capacity without adding more beds to the hospital. Additionally, this eases up beds for emergency surgeries that can’t be rescheduled. Note that this study is not aim to change how surgical blocks are allocated to the surgeons or how they are currently being used, we simply rearrange the distribution of blocks in the days of the week [2].

## 2 Project Status and Differences

In our initial project plan, we aimed to replicate the entire process in the paper “*An Optimization Framework for Smoothing Surgical Bed Census via Strategic Block Scheduling*” [2]. However, we encountered troubles when finding similar dataset used in that paper, so we had to create our own dataset for this project. All of the variables introduced in the paper [2] are also used in our study, apart from the variable  $F$  which represents the surgical floor units. We still defined this variable in our code but decided not to use it eventually in order to avoid heavy computation. People who wish to put this variable in use can do it easily by amending our code.

### 2.1 Constraint Changes

From the paper [2], we found an error in the model they used (or we simply could not understand it). In their model, they allowed a single block to be moved to different days. For example, in their model, a block on Tuesday can be moved to both Wednesday and Friday. This

does not make sense to us because one block cannot exist in two days naturally. Therefore, we introduced an additional hard constraint (2) that forbids this kind of move, as can be seen in the Model Development section.

## 2.2 Data Generation

As mentioned, one of the first road blocks we had was finding suitable data. Carnes uses secure data from the Massachusetts General Hospital [2]. After not finding suitable data for the first couple weeks, we followed our initial plan and began generating our own data. Carnes included some basic figures (weekday bed counts, number of OR rooms, bed capacity) in the paper, from these figures we can generate the sets used by the integer programming model in the paper [2]. Using Figure 8 from the paper, we can create initial bed capacity for a preexisting block assignment. To generate the preexisting block assignment, we assigned the blocks on the first day to 52 random surgeons. Repeating this for 1 month (28 days), we constructed a data set the model can use. Then assigning surgeons to each service in blocks of 20 we have our initial block assignment.

For the model formulation given in the paper [2], the set  $P$  is potentially ambiguous.  $P$  is defined as the pair of blocks corresponding to allowed moves. This set could include many constraints that are not discussed in the paper. From Figure 8 in the paper [2], we see there is some constraint on moving blocks to weekends, as even after the permutation weekends remain the lowest points in the week. For these reasons we will test our model on a variety of  $P$ . We create multiple  $P$  to include the following constraints:

- **Weekend Constraint:** Only being able to move blocks on weekends to other weekend days or Fridays.
- **Surgical Chiefs:** Including surgical chiefs, meaning that we remove block pairs from  $P$  if they include a block owned by a surgical chief.
  - We assume there are same number of surgical chiefs in each service.
- **Close Rooms:** Only allowing block moves if the rooms are within 10 of each other.
  - This drastically reduces the number of allowed block moves, which greatly helps computational time, but doesn't significantly affect the results.

## 2.3 Model Development

Following the ideas in the paper “*An Optimization Framework for Smoothing Surgical Bed Census via Strategic Block Scheduling*” [2] and also the constraints mentioned in 2.1, we build the following model:

$$\max \sum_{f \in F} (peak_f - z_f) \quad (1)$$

$$\text{s.t.} \quad \sum_{\substack{b_2 \in B: \\ (b_1, b_2) \in P}} x_{b_1 b_2} = 1 \quad \forall b_1 \in B \quad (2)$$

$$\sum_{\substack{b_1 \in B: \\ (b_1, b_2) \in P}} x_{b_1 b_2} = 1 \quad \forall b_2 \in B \quad (3)$$

$$\sum_{\substack{(b_1, b_2) \in P: \\ b_2 = (o_2, d_2), \\ i = d - d_2 \pmod{7}}} cases_{fb_1 i} \cdot x_{b_1 b_2} \leq z_f \quad \forall d \in W, f \in F \quad (4)$$

$$\sum_{\substack{(b_1, b_2) \in P: \\ b_2 = (o_2, d), \\ (b_1, i) \in D_s}} x_{b_1 b_2} \leq 1 \quad \forall d \in W, i \in N, s \in S \quad (5)$$

$$\ell_v \leq \sum_{\substack{(b_1, b_2) \in P: \\ b_2 = (o_2, d)}} alloc_{b_1 v} \cdot x_{b_1 b_2} \leq u_v \quad \forall d \in W, v \in V \quad (6)$$

## 2.4 The Thursday Constraint

The paper [2] includes a constraint which is concerned with moving blocks in Thursday, where all blocks in Thursday have to be moved as a group. We implement this feature by adding an additional binary decision variable  $t_d$  for  $d \in W$  which indicates whether Thursday blocks are moved to weekday  $d$ . Therefore, we implement the Thursday constraint by adding the following constraint in the model:

$$\sum_{\substack{(b_1, b_2) \in P: \\ b_1 = (o_1, 4), \\ b_2 = (o_2, d)}} x_{b_1 b_2} \leq t_d \quad \forall d \in W$$

Since all Thursday blocks have to move to the same day, we also need to constrain the variable  $t_d$  using:

$$\sum_{d \in W} t_d = 1.$$

Adding this into our model gives a permutation illustrated in Figure 1, which does not reflect the results in the paper (see Figure 8). It appears to simply permute the Thursday and Monday block allocations with little improvement in reducing the peak of the occupied beds. For this reason, we decided to proceed without including this constraint.

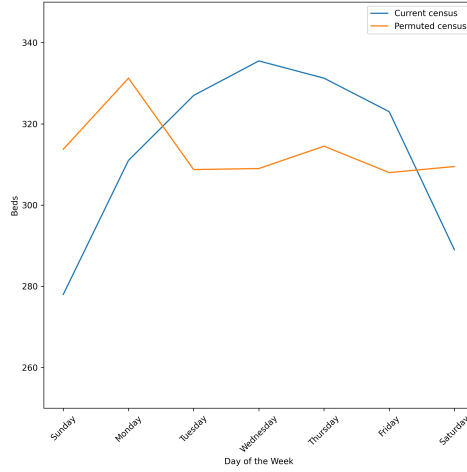


Figure 1: Model with Thursday Constraint

### 3 Results

In this section we will run our model on different sets of  $P$  outlined in section 2.2. We will first run a baseline model, then implement additional constraints discussed in Section 2.2. Note that for all the following results we included the **close rooms** constraint (mentioned in Section 2.2) to reduce the computational time.

#### 3.1 The Baseline Model

Running our current model detailed in Section 2.3, where the set  $P$  has only been restricted by the room constraint gives us the following result,

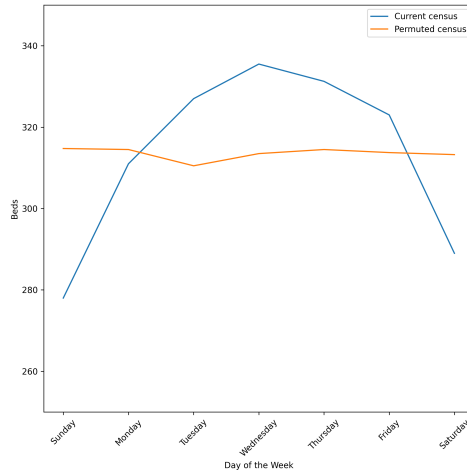


Figure 2: Baseline model

With no additional constraints we see the model gives us a census close to a flat line. This is not representative of real applications and of the results seen from the paper [2]. However it indicates that our model is performing as we would expect, successfully reducing the peak.

#### 3.2 The Weekend Constraint

After applying the **weekend** constraint from 2.2, we received the following permutation:

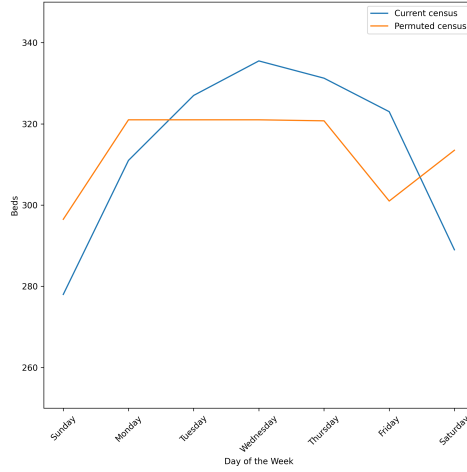


Figure 3: Permutation with the ‘weekend’ constraint

By comparing the updated permutation with the baseline plot, we can find that the bed occupancy on Friday drops remarkably, and it becomes almost constant from Monday to Thursday. Applying this constraint made the permuted census on Monday exceed the current census, but it resolves the high bed occupancy on Friday and weekends. Additionally, including the weekend constraint follows trends we see from the paper’s [2] results (Figure 8), as even after the block moves the weekends have the lowest bed occupancy.

### 3.3 Surgical Chief Constraint

We are also interested in the impact that the number of surgical chiefs (mentioned in 2.2) has on bed occupancies. Firstly, we set the number of surgical chiefs to 5, leading to the plot:

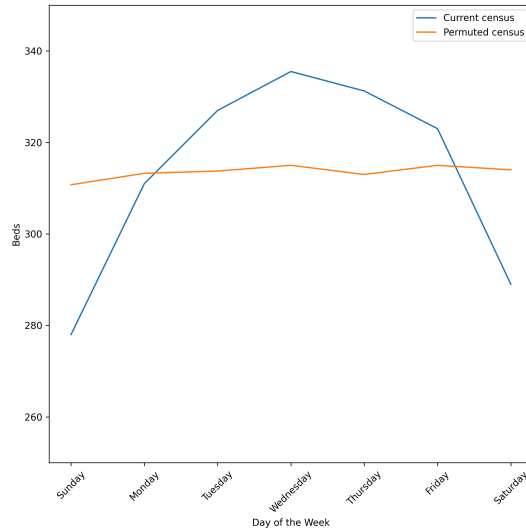
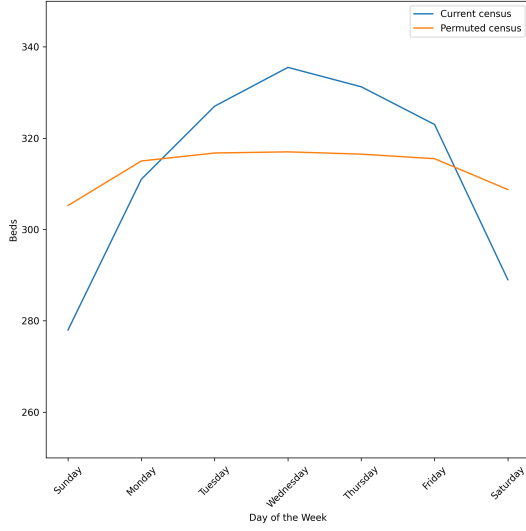
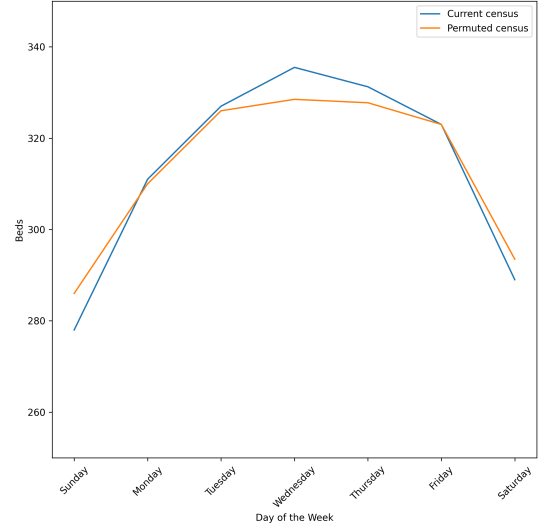


Figure 4: Permutation of the surgical chief constraints with 5 chiefs



(a) 10 Chiefs



(b) 20 Chiefs

Figure 5: Permutation of the surgical chief constraints with more chiefs

Figure 4 is highly similar to the baseline result and it shows the permuted census is almost flat across different days of the week. As we increase the number of surgical chiefs in the model, the permutation is expected to align more closely with the current census. This is intuitive because a higher number of chiefs reduces the number of blocks that are allowed to move.

### 3.4 Weekend Constraint with Surgical Chief Constraint

When applying both the weekend constraint and the surgical chief constraint (with 5 chiefs), we obtain the result detailed in Figure 6:

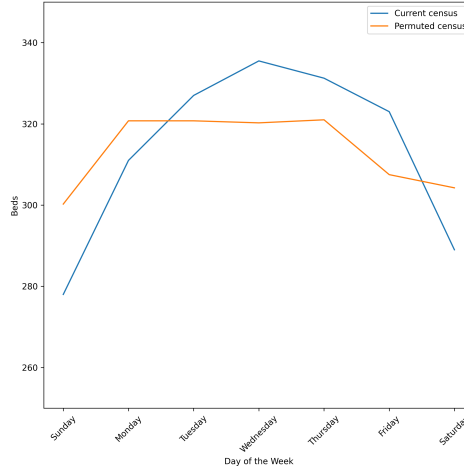


Figure 6: Permutation with both constraints

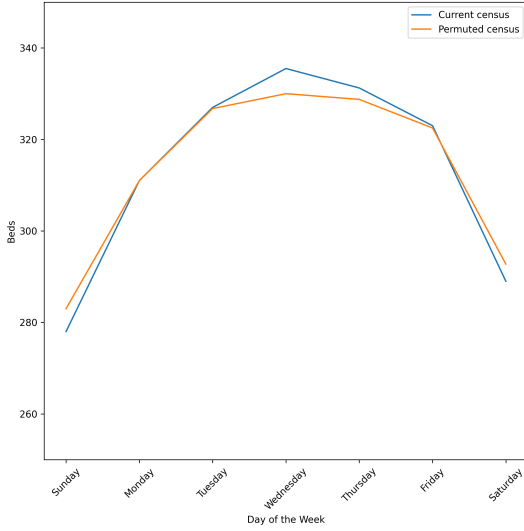
Compared to the results of the surgical chief constraints in Figure 4 and the weekend constraints in Figure 3, the permutation falls between the two. This suggests that although assigning five chiefs alone may not significantly change the outcome, it can influence the behavior when combined with other constraints. This is our final result in replicating the findings in the paper [2]. It takes into account the main constraints we have discussed and applies a similar formulation as the paper [2].

## 4 Additional Features

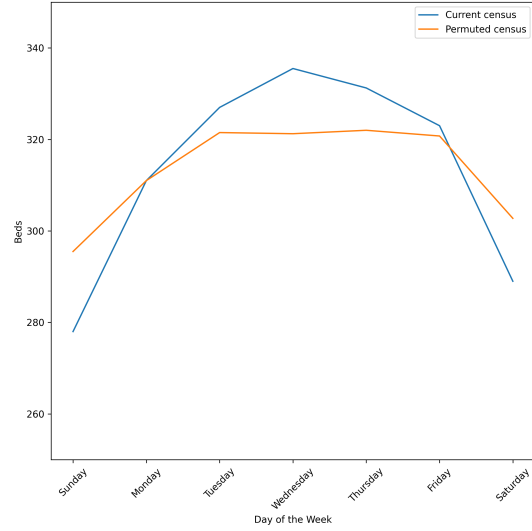
In this section we will be expanding the work covered in the paper [2]. Given we were working with a preexisting schedule and people prefer to keep consistent work hours, we implemented a cost term  $\alpha$  into our objective function,

$$\max \sum_{f \in F} (peak_f - z_f) + \alpha \sum_{\substack{(b_1, b_2) \in P: \\ b_1 = b_2}} x_p.$$

This penalizes us for every move we make whilst still aiming to reduce the peak bed occupancy. Adjusting  $\alpha$  to 0.4 we find the following bed census in Figure 7a,



(a) Optimal permutation with  $\alpha = 0.4$ .



(b) Optimal permutation with  $\alpha = 0.2$

Figure 7: Comparison of Permutation: with different  $\alpha$  values

As expected the peak is not reduced as much compared to our previous findings, as we only have 17 block moves. However, by reducing the number of block moves we have dramatically reduced the variability of our surgeon's schedules. Now adjusting  $\alpha$  to 0.2 we obtain the bed census shown in 7b. As expected for smaller values of  $\alpha$  we obtain a flatter solution. This comes at the cost of increasing block moves and variability of surgeons' schedules, as now we have 48 moves over the whole month.

## 5 Discussion

In conclusion, our project replicated the essential parts of the process mentioned in paper [2], including the Thursday Constraint, the Weekend Constraint and the Surgical Chief constraint. We compared the bed occupancies throughout a week based on different constraints applied. Through experiments and implementations, the Thursday Constraint is excluded as it failed to illustrate the expected results in the paper, while the other two constraints that were successfully implemented works for the objective of smoothing surgical bed census. And improving upon the MIP formulation given in the paper [2], we introduced a new constraint to not allow blocks to be moved to different days.

As well as replicating the results from the paper [2], we implemented a new variable  $\alpha$ , the cost term to penalizes the model for every block move it makes, as described in section 4. Although

it is not mentioned in the paper [2], we found that a constraint that limits the block movement on weekends might exist as there is an obvious drop of bed occupancy during weekends in their paper.

We must also acknowledge that there are some limitations in this study. First of all, we failed to replicate all the works in the paper [2] because we could not find the same dataset they used. Moreover, some of the constraints in their model are not explained. For example, a constraint to account for the interaction between ORs was used, but the paper [2] was not specific on how these blocks were linked (i.e. linking by day or by room number). Due to this, we removed it from our formulation which resulted in differences in our model. Future study could explore in more detail the interaction constraint and how this affects the bed census. Secondly, in this study we only considered one floor ( $F = \{1\}$ ), due to computational requirements. However, future work can easily expand our code to include multiple floors allowing for more complex scenarios. Finally, we implemented a surgical chief constraint as mentioned in section 3.4. Due to differences in hospital operation, there is no standard for the number of required surgical chiefs. For this reason we ran our model for varying numbers of chiefs, to show the impact this can have on peak reduction. Through this report we have shown the versatility and effectiveness of our block scheduling formulation in order to reduce peak bed census.

## 6 Appendix

### 6.1 Code Access

The dataset and the code of this project can be accessed via GitHub

### 6.2 Plots from “An Optimization Framework for Smoothing Surgical Bed Census via Strategic Block Scheduling” [2]

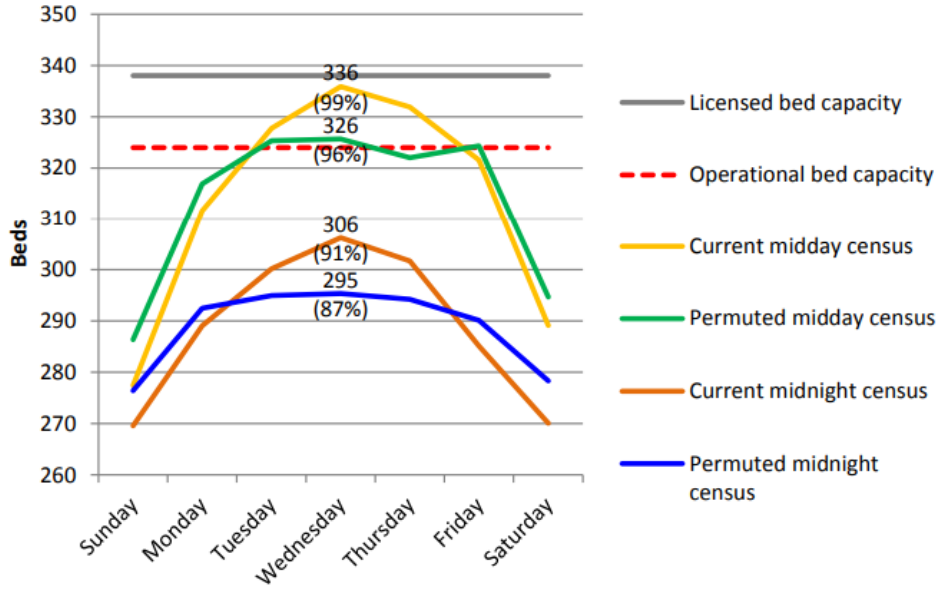


Figure 8: Results before and after the permutation from the paper [2]

## References

1. Rais, A. & Viana, A. Operations research in healthcare: a survey. *International transactions in operational research* **18**, 1–31 (2011).
2. Carnes, T. *et al.* An optimization framework for smoothing surgical bed census via strategic block scheduling. *Manufacturing Service Operation Management*, 488–494 (2011).